

Hrishitaa Dharmavarapu

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EDUCATION

University of Illinois at Chicago

Masters in Computer Science

- Coursework: Agentic AI, Advanced Machine Learning, Computer Algorithms, Socially Responsible AI.

Chicago, IL

Aug. 2024 – May 2026

KL University

Bachelor of Technology in Computer Science

- Coursework: DBMS, Distributed Systems, Data Structures and Algorithms, System Design and Patterns.

Hyderabad, India

June 2019 – Mar 2023

EXPERIENCE

Graduate Research Assistant, UIC Business School

University of Illinois at Chicago

- Architected a scalable manager-worker multi-agent system with LangChain and Amazon Bedrock, delivering structured extraction from unstructured PDFs across 100+ municipalities at 95%+ accuracy
- Engineered a Chroma RAG pipeline with prompt design and context management, lifting retrieval precision by 40% across inconsistent document formats
- Instrumented the pipeline with Datadog telemetry and automated failure detection, cutting silent data corruption by 70%
- Presented the work to the Sustainability Research and Innovation Congress, reaching an audience across 80+ countries

May 2025 – May 2026

Chicago, IL

Full Stack Developer

University of Illinois at Chicago

- Built and shipped a full-stack mental health application with React, FastAPI, and PostgreSQL, leveraging Cursor AI to cut development time by 50%
- Deployed to production on AWS with CI/CD and monitoring, now used internally across UI Health after piloting with 15 teams

Oct 2024 – May 2025

Chicago, IL

Associate Software Engineer

OpenText Technologies

- Designed a React and TypeScript frontend over a FastAPI backend using modular object-oriented design, delivering real-time dashboards to enterprise clients at scale
- Architected and scaled backend services behind an AWS API Gateway, automating build, test, and deployment via CI/CD to cut release time by 40%
- Used Datadog APM to pinpoint SQL bottlenecks on a 2M+ user platform, applying query tuning and indexing to cut response times by 30%
- Built automated end-to-end validation and UI test suites, driving up automation coverage and cutting manual testing effort by 60%

Oct. 2022 – June 2024

Hyderabad, India

Software Engineering Intern

OpenText Technologies

- Owned resolution of a production-breaking 1 TB upload failure on a 2M+ user platform, implementing S3 multipart uploads to slash upload time by 60%
- Drove backend reliability with functional and unit test suites in Pytest, reducing production defects by 30%

Aug. 2022 – Oct. 2022

Hyderabad, India

PROJECTS

Clinical Documentation Helper | *LangGraph, AWS Textract, React, FastAPI, PostgreSQL*

May 2025 – May 2026

- Built a multi-agent GenAI system with LangGraph and AWS Textract that extracts data from clinical document images, maintains patient records, and automates scheduling, checkup reminders, and patient notifications
- Added guardrails with PII redaction and human-in-the-loop approval before critical actions, and shipped a React, FastAPI, and PostgreSQL platform now used internally by UI Health clinicians [\[Link\]](#)

Behavioral Health Reference Matcher | *LangGraph, RAG, ChromaDB, FastAPI, React, PostgreSQL*

Jan. 2026

- Built a staff-facing referral tool that parses free-text client needs and matches them to treatment programs using hybrid retrieval, combining structured filters with semantic search over ChromaDB
- Designed a LangGraph agent with input and output hooks that parses requirements, ranks programs, and explains each match, achieving 85% recall@5 on a 15-query evaluation set [\[Link\]](#)

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, HTML, CSS, SQL, Bash

AI/ML Engineering: LLM APIs, LangGraph, RAG, ChromaDB, Prompt Engineering, Context Management, Agent Evaluation, Guardrails, PII Redaction, Human-in-the-Loop, Hooks, LangSmith

Full-Stack & API Development: React, REST APIs, gRPC, GraphQL, FastAPI, Flask, Webhooks, Stripe, UI/API/DB Architecture

Data & Search: Spark, Hadoop, Elasticsearch, Redis, PostgreSQL, pandas, NumPy, Matplotlib

Cloud, DevOps & Tools: AWS, Vercel, API Gateway, S3, Docker, Kubernetes, CI/CD, A/B Testing, Datadog, Pytest, Git, Cursor AI, Claude Code, Salesforce, Jira